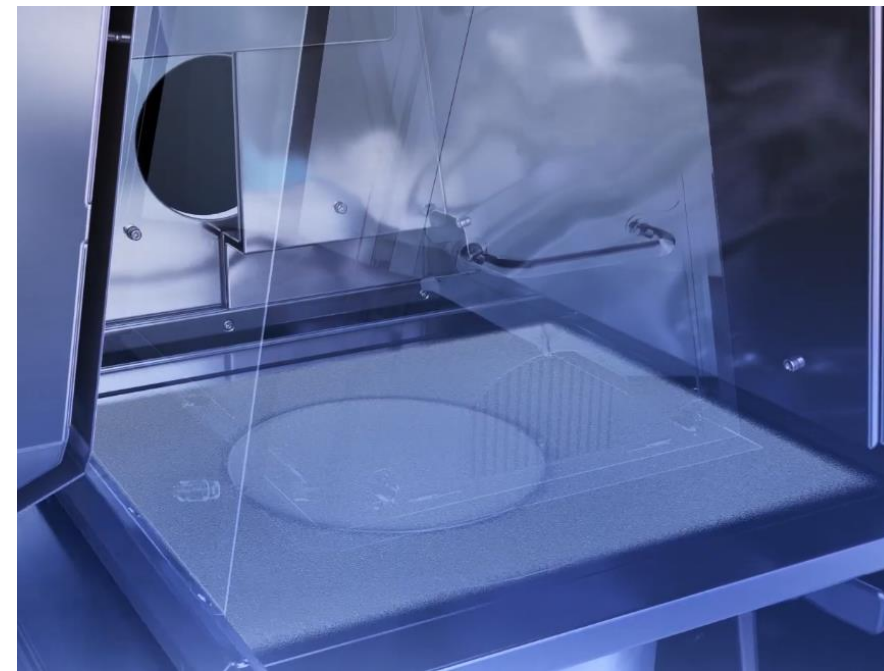
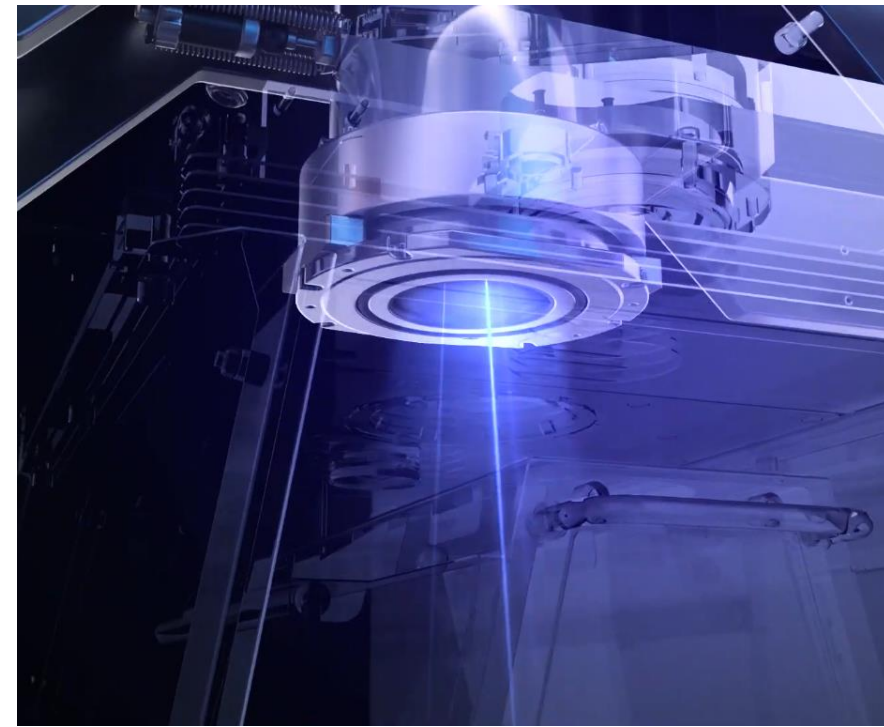


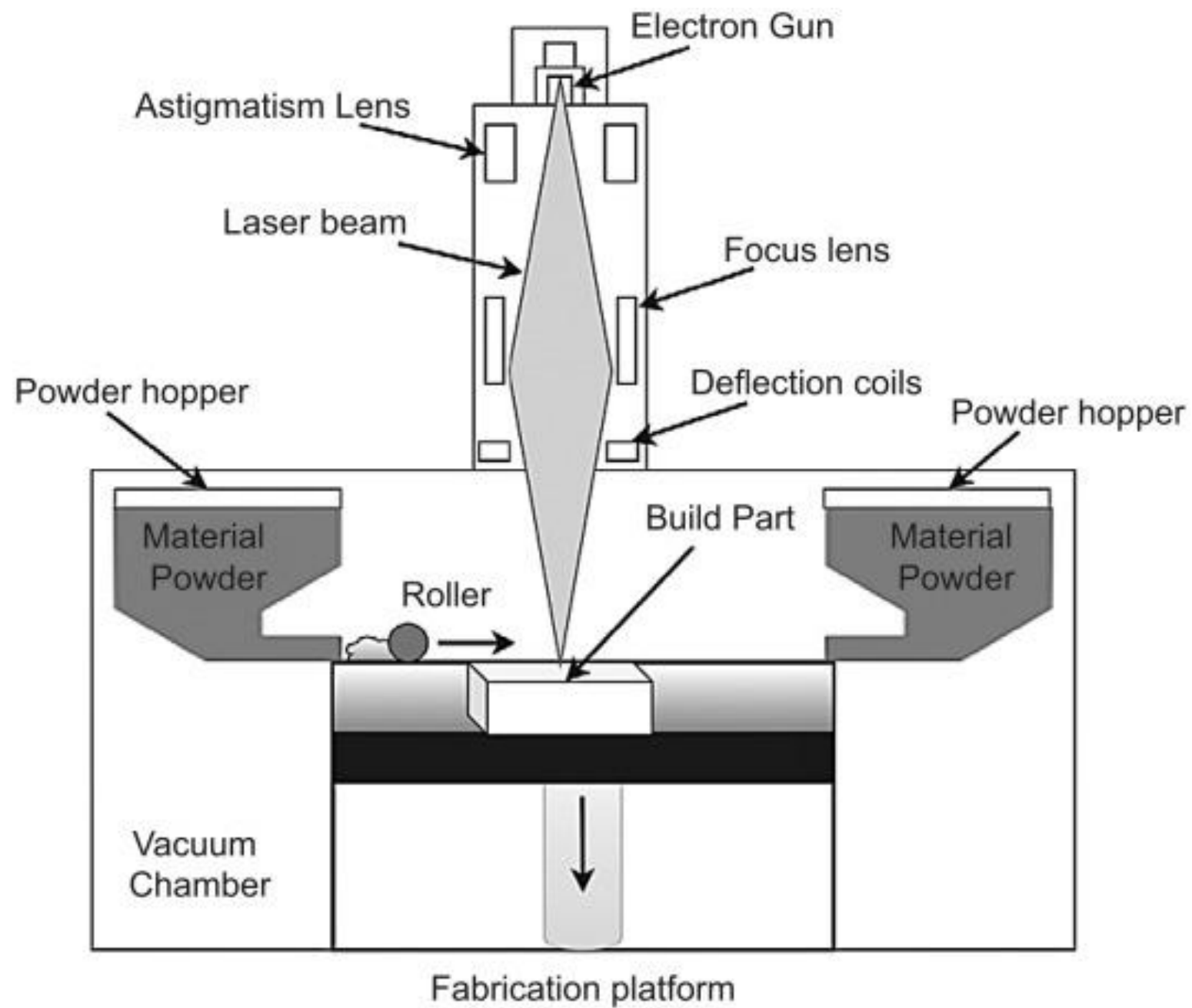
Electron beam melting

- Arcam AB, a Swedish technology development company was founded in 1997.
- The company's main activity is concentrated in the development of the Electron Beam Melting (EBM) technique for the production of solid metal parts directly from metal powder based on a 3D CAD model.
- The fundamental development work for Arcam's technology began in 1995 in collaboration with Chalmers University of Technology in Gothenburg, Sweden.
- The Arcam EBM technology was commercialized in 2001.
- The company address is at Krokslatts Fabriker 30, SE-431 37 Molndal, Sweden.

Principle

- Parts are built up when an electron beam is fired at the metal powder. The computer controlled electron beam in vacuum melts the layer of powder precisely as indicated by the CAD model with the gain of the electrons' kinetic energy.
- The building of the part is accomplished layer by layer. A layer is added once the previous layer has melted. In this way, the solid details are built up of thin metal slices melted together.



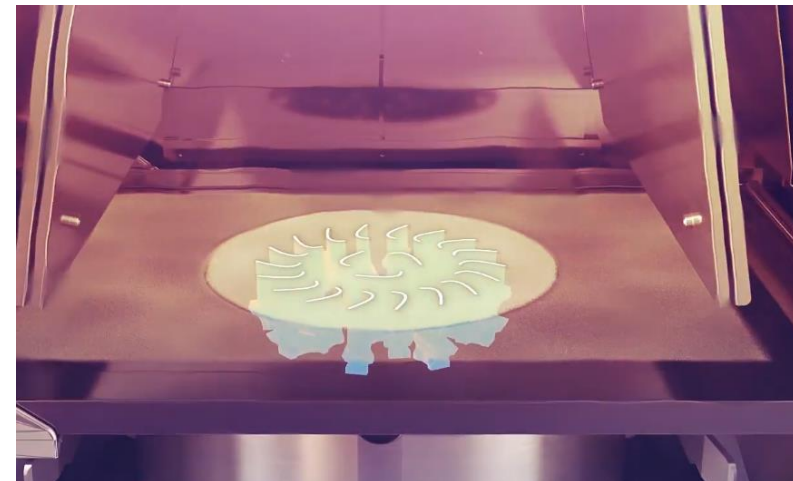
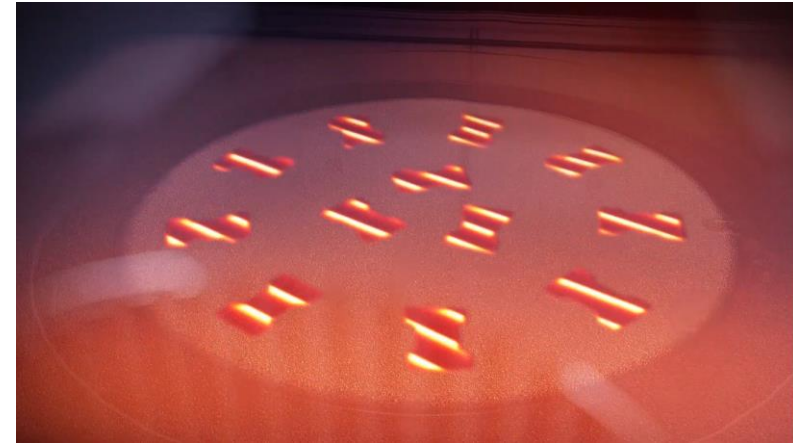


Process

(1) The part to be produced is first designed in a **3D CAD** program. The model is then sliced into thin layers, approximately a tenth of a millimeter thick.

(2) An equally **thin layer of powder** is scraped onto a vertically adjustable surface. The first layer's geometry is then created through the layer of powder melting together at those points directed from the CAD file with **a computer-controlled electron beam**.

(3) Thereafter, the building surface is lowered and the next layer of powder is placed on top of the previous layer. The procedure is then repeated so that the object from the CAD model is shaped layer by layer until a finished metal part is completed.



ADVANTAGES OF EBM

- The higher energy source makes it possible to use larger diameter metal powder.
- That heavier powder metal is easier to work with, safer to work with.
- Electron beam melting takes place at higher temperatures than laser powder bed fusion, which means you have better management over the **thermal stresses**.
- That **means less warping, less distortion, better dimensional accuracy**.
- And it also means that post-processing steps like hot isostatic pressing or HIPping probably aren't going to be necessary.
- The control over thermal stresses means that it's **not necessary to rely on support structures** to anchor the part in place against deformation.
- the build rate is just faster

DISADVANTAGES OF EBM

- Cooling takes place inside the machine. Can take a few hours. Powder removal can be a challenge.
- powder trapped inside that can be difficult to get out if larger powder size is used
- So might need additional post-processing steps, things like ultrasonic powder removal.
- might need to add extra inspection into your process just to be absolutely sure there's no unmelted powder inside.
- Compared to other powder bed fusion, Electron beam melting produces a **coarser surface**.
- Laser powder bed fusion has an advantage when it comes to very fine precision work.
- But both of these processes produce parts that are very close to **net shape**.

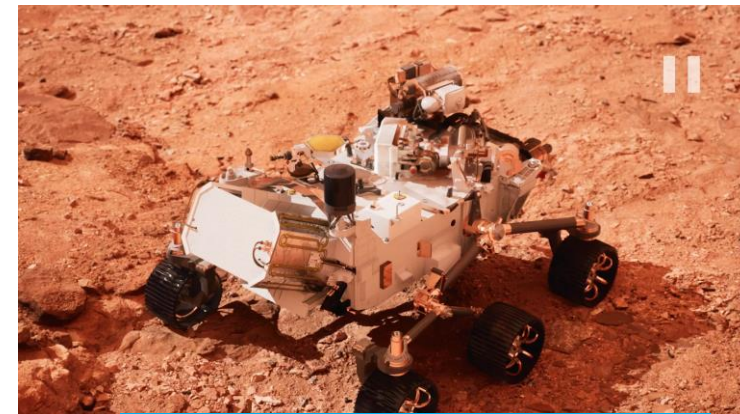


coarser surface.

Laser powder bed fusion = selective laser sintering

Applications

- electron beam melting used in all kinds of industries, everything from aerospace and energy to medical devices.
- it build turbine blades, satellite brackets, medical implants, even parts for the perseverance Mars rover.
- Electron beam melting is very commonly used to make medical implants like hip cups where an engineered rough surface is actually beneficial to the patient's healing.



perseverance Mars rover.